

Electronegativity & Bonding of the Period 3 Elements (Cambridge (CIE) A Level Chemistry): Revision Note

Trends in Period 3 Electronegativity & Bonding

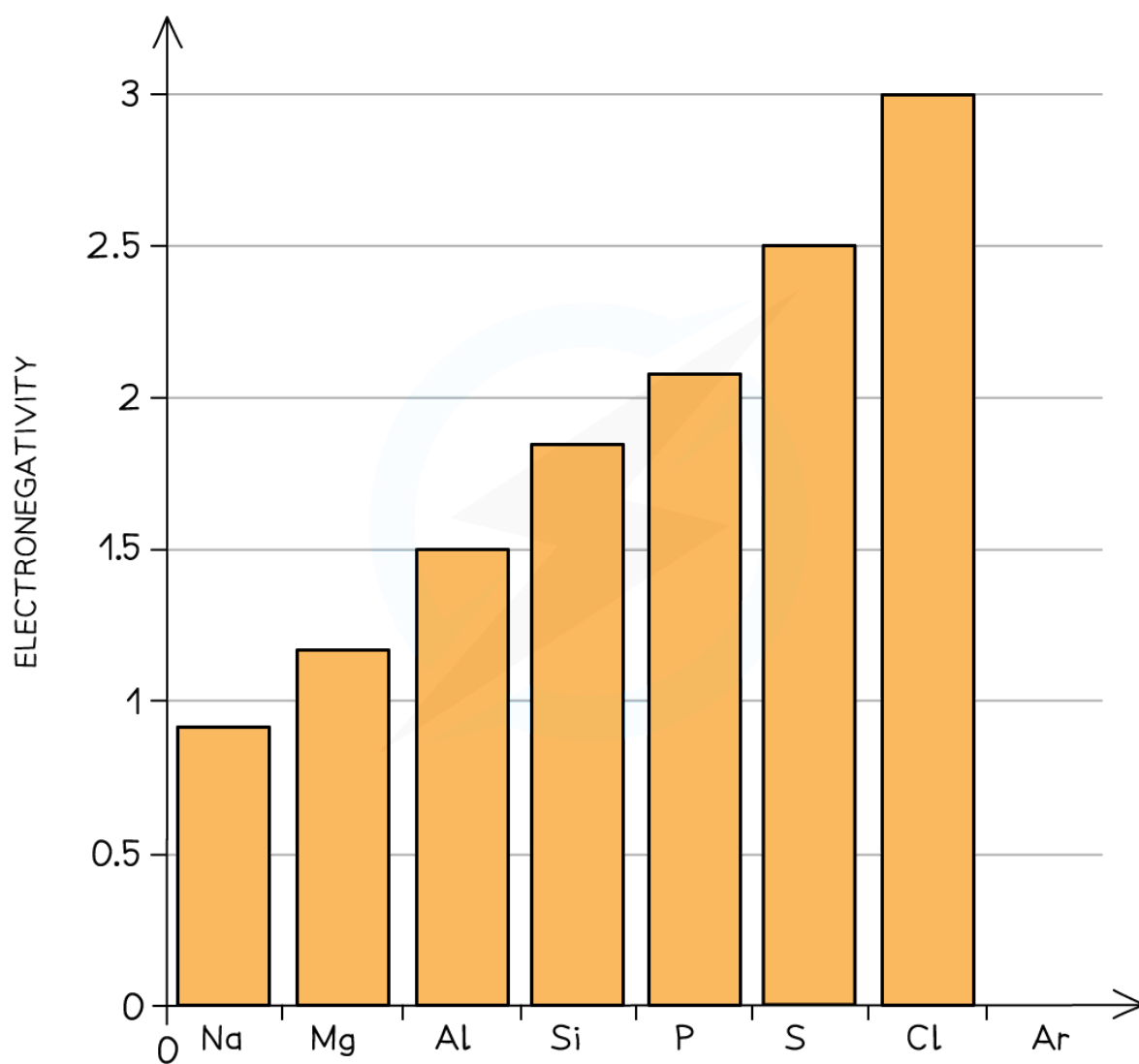
Electronegativity

- **Electronegativity** is the power of an element to draw the electrons towards itself in a covalent bond
- Going **across** the period, the **electronegativity** of the elements increases

Electronegativity across Period 3 table

Period 3 element	Na	Mg	Al	Si	P	S	Cl	Ar
Electronegativity	0.9	1.2	1.5	1.8	2.1	2.5	3.0	-

Graph of the electronegativity of the Period 3 elements



Copyright © Save My Exams. All Rights Reserved

Electronegativity of the Period 3 elements increases from Na to Cl

- As the **atomic number** increases going across the period, there is an **increase** in **nuclear charge**
- Across the period, there is an increase in the number of **valence electrons** however the **shielding** is still the same as each extra electron enters the same shell
- As a result of this, electrons will be more strongly attracted to the nucleus causing an increase in electronegativity across the period

Bonding & structure of Period 3 elements table

Period 3 element	Na	Mg	Al	Si	P	S	Cl

Bonding	Metallic	Metallic	Metallic	Covalent	Covalent	Covalent	Covale
Structure	Giant metallic	Giant metallic	Giant metallic	Giant molecular	Simple molecular	Simple molecular	Simpl molecu

Bonding and structure from Al to S

- As you move across the Periodic Table from aluminium (Al) to sulfur (S), both bonding and structure change:
 - Bonding changes from **metallic** (in Al) **to covalent** (in Si, P, S, etc.)
 - Structure changes from **giant lattices to simple molecular structures**
- Sodium (Na), magnesium (Mg), and aluminium (Al) are all metals
- They form a **giant metallic lattice**:
 - Positive metal ions are arranged in a regular lattice
 - They are surrounded by a 'sea' of delocalised electrons
 - These electrons come from the outer shell (valence shell) of each atom

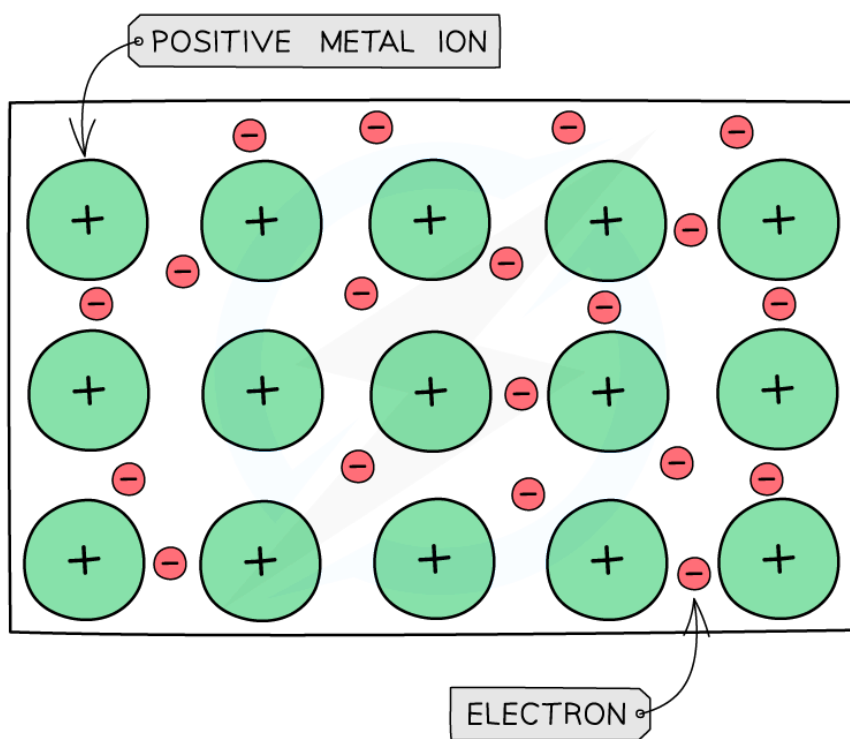
Delocalised electrons and bond strength

- Na donates **1** electron per atom
- Mg donates **2** electrons per atom
- Al donates **3** electrons per atom
- As a result:
 - More **delocalised electrons = stronger electrostatic forces** between the metal ions and the electron cloud
 - Al^{3+} forms stronger metallic bonds than Na^+ , due to:
 - Higher ionic charge
 - Greater number of delocalised electrons

Electrical conductivity

- Because aluminium contributes more delocalised electrons, it has:
 - More **charge carriers**
 - Stronger **metallic bonding**
 - Therefore, aluminium is a **better conductor** of electricity than sodium or magnesium

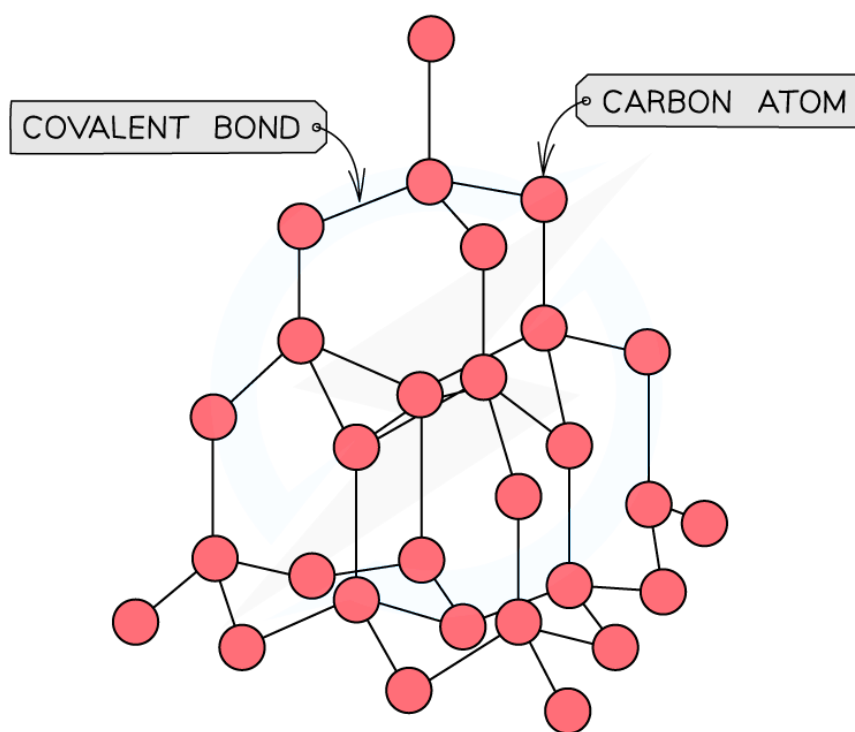
A giant metallic lattice



Metal cations form a giant lattice held together by electrons that can freely move around

- Si is a non-metallic element and has a **giant molecular** structure in which each Si atom is held to its neighbouring Si atoms by **strong covalent bonds**
- There are no delocalised electrons in the structure of Si which is why silicon cannot conduct electricity and is classified as a **metalloid**

The giant molecular structure of silicon



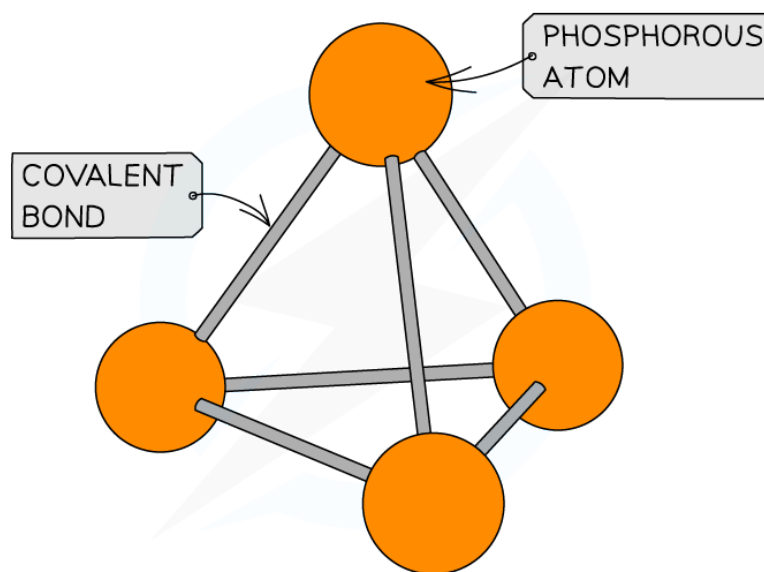
Copyright © Save My Exams. All Rights Reserved

The diagram shows the giant molecular structure of silicon where silicon atoms are held together by strong covalent bonds

Bonding from P to Ar

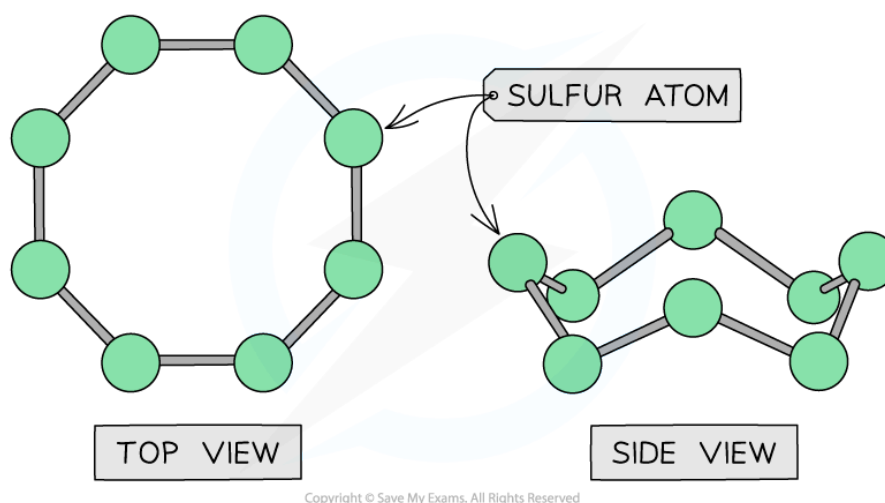
- **Phosphorous, sulfur, chlorine** and **argon** are non-metallic elements
 - Phosphorous, sulfur and chlorine exist as simple molecules (P_4 , S_8 , Cl_2)
 - Argon exists as **single** atoms
- The **covalent bonds within** the molecules are strong, however, **between** the molecules there are only weak **instantaneous dipole-induced dipole forces**
- It doesn't take much energy to break these **intermolecular** forces
- The lack of delocalised electrons means that these compounds cannot conduct electricity

The simple molecular structure of phosphorous



The diagram shows the simple molecular structure of phosphorus with covalent bonds between the atoms

The simple molecular structure of sulfur



The diagram shows the simple molecular structure of sulfur with covalent bonds between the atoms